

Yunkee Chae

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Personal Data

Address Address 18-211, 1 Gwanak-ro, Gwanak-gu, Seoul (Postal code: 08827)
Birth 19th Sep, 1995, in Republic of Korea
Nationality Korea, Republic of

Education

Seoul National University Mar 2014 – Feb 2021
Bachelor of Science in Mathematical Sciences
Supervisor: Prof. David Donghoon Hyeon

Seoul National University Mar 2021 – Aug 2026 (Expected)
Master's and Ph.D. Integrated Course of Engineering
Interdisciplinary Program in Artificial Intelligence
Music and Audio Research Group (MARG)
Supervisor: Prof. Kyogu Lee
Thesis: Unified Latent Diffusion Models for Simultaneous Multi-Track Music Generation and Separation

Research Interest

Deep learning / Machine learning

Audio generation

- Multi-track music generation

Audio Enhancement / Restoration

- Live performance audio enhancement

Source Separation

- Query-based source separation
- Source separation via generative modeling

Audio Compression

- Variable Bitrate Neural Audio Codec

Publications

First Author

MGE-LDM: Joint Latent Diffusion for Simultaneous Music Generation and Source Extraction 2025
Yunkee Chae and Kyogu Lee

Accepted to Advances in Neural Information Processing Systems (NeurIPS 2025)

- Developed a unified latent diffusion framework (MGE-LDM) that enables simultaneous music generation, source imputation, and text-driven source extraction via conditional inpainting, without relying on fixed instrument classes.
- Pioneered the use of inpainting in latent diffusion for source extraction, supporting flexible, class-agnostic manipulation across diverse multi-track datasets (e.g., Slakh2100, MUSDB18, MoisesDB).
- [[Paper](#)] [[Samples](#)] [[Code](#)]

Towards Bitrate-Efficient and Noise-Robust Speech Coding with Variable Bitrate RVQ 2025
Yunkee Chae and Kyogu Lee

Accepted to Proceedings of the INTERSPEECH 2025

- Introduced a dynamic bitrate adjustment framework optimizing rate-distortion trade-offs by prioritizing critical speech components and suppressing noise, outperforming constant bitrate RVQ.
- Integrated feature denoiser to enhance robustness, achieving superior compression efficiency and perceptual quality in noisy environments.
- [[Paper](#)] [[Samples](#)] [[Code](#)]

Song Form-aware Full-Song Text-to-Lyrics Generation with Multi-Level Granularity

2025

[Yunkee Chae](#) , Eunsik Shin, Suntae Hwang, Seungryeol Paik, and Kyogu Lee

Accepted to Proceedings of the INTERSPEECH 2025

- Proposed a method for generating lyrics with precise control over syllable counts at the paragraph, line, phrase, and word levels, conditioned on a text prompt and song form structure.
- First work to generate comprehensive lyrics with song forms and syllable conditions.
- [[Paper](#)] [[Samples](#)] [[Demo](#)]

Variable Bitrate Residual Vector Quantization for Audio Coding

2024

[Yunkee Chae](#) , Woosung Choi, Yuhta Takida, Junghyun Koo, Yukara Ikemiya, Zhi Zhong, Kin Wai Cheuk, Marco A. Martínez-Ramírez, Kyogu Lee, Wei-Hsiang Liao, and Yuki Mitsufuji

Accepted to Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2025) (Oral presentation)

Accepted to NeurIPS 2024 Workshop on Machine Learning and Compression

- The first work to propose training framework for variable bitrate compression within the residual vector quantization.
- Improved the rate-distortion tradeoff performance for state-of-the-art RVQ codecs.
- Work done during the internship at Sony AI, Tokyo.
- [[Paper \(ICASSP ver.\)](#)] [[Paper \(NeurIPS workshop ver.\)](#)] [[Samples](#)] [[Code](#)]

Exploiting Time-Frequency Conformers for Music Audio Enhancement

2023

[Yunkee Chae](#) , Junghyun Koo, Sungho Lee, and Kyogu Lee

Accepted to Proceedings of the 31th ACM International Conference on Multimedia (ACM MM 2023)

- Proposed a system for enhancing degraded musical recordings based on the Conformer architecture.
- Capable of perform music enhancement with multi-track mixtures, which has not been examined in previous work.
- [[Paper](#)] [[Samples](#)] [[Code](#)]

Co-First Author

Self-refining of Pseudo Labels for Music Source Separation with Noisy Labeled Data

2023

Junghyun Koo*, [Yunkee Chae](#) *, Chang-Bin Jeon, and Kyogu Lee

Accepted to Proceedings of the 24nd International Society for Music Information Retrieval Conference (ISMIR 2023)

- Introduced an automated technique for refining the labels in partially mislabeled dataset.
- Proposed a training framework for music source separation with multi-labeled data.
- Participated in Sound Demixing Challenge 2023, Music Demixing Track A ([link](#))
- [[Paper](#)]

Debiased Automatic Speech Recognition for Dysarthric Speech via Sample Reweighting with Sample Affinity Test

2023

Eungbeom Kim*, [Yunkee Chae](#) *, Jaeheon Sim, and Kyogu Lee

Accepted to Proceedings of the INTERSPEECH 2023

- Proposed an automatic speech recognition system robust for dysarthric speech.
- Introduced novel sample reweighting method called sample affinity test.
- [[Paper](#)]

Co-Author

Show Me the Instruments: Musical Instrument Retrieval from Mixture Audio

2023

Kyungsu Kim*, Minju Park*, Haesun Joung*, [Yunkee Chae](#) , Yeongbeom Hong, Seonghyeon Go, and Kyogu Lee

Accepted to Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2023)

- Proposed a novel method for musical instrument retrieval system for mixture audio using single-instrument encoder and the multi-instrument encoder
- [[Paper](#)] [[Samples](#)] [[Code](#)]

Preprint

InterStem: Joint Variable-Stem Modeling for Simultaneous Music Generation and Separation

2026

[Yunkee Chae](#) , Christoph Boeddeker, Yoshiki Masuyama, Julius Richter, Gordon Wichern and Jonathan Le Roux

Submitted to Proceedings of the INTERSPEECH 2026 long paper track

- under review.
- Work done during the internship at MERL
- Proposed a multi-track diffusion modeling with variable number of stems.
- Support stem generation and separation in single diffusion framework.
- [[Paper](#)]

Yunkee Chae and Kyogu Lee

The First Korean Society for Music Informatics Conference (KSMI), Extended Abstract track (oral)

- Proposed an open-set source separation framework using mixture-conditioned latent diffusion, without assuming a fixed stem taxonomy or known source count.
- Demonstrated that stochastic reverse sampling from a single mixture-conditioned diffusion model yields multi-modal outputs clustering around constituent sources in the CLAP embedding space.
- [[Paper](#)] [[Slides](#)]

(* equal contribution)

Work / Research Experience

Research ScientistMay 2026 – present

at Moises.ai

Research InternAug 2025 – Feb 2026

at MERL (Mitsubishi Electric Research Laboratories), Cambridge, MA, USA

- Research on simultaneous multi-track music audio generation and source separation

Research InternMay 2024 – Sep 2024

at Sony AI, Tokyo, Japan

- Variable bitrate residual vector quantization for audio coding
- Paper accepted at ICASSP 2025 & NeurIPS 2024 Machine Learning and Compression Workshop

Research InternJan 2021 – Feb 2021

at Music and Audio Research Group (MARG), Seoul National University

- Source separation of individual guitar strings from polyphonic guitar recordings

Software EngineerSep 2020 – June 2021

at Hmachines Inc., Seoul, Korea

- 3D reconstruction of smooth surface via laser scanner using multiple view geometry

Research InternMar 2020 – Sep 2020

at Department of Mathematical Science, Seoul National University

- 3D reconstruction of projected grids via projector calibration
- Supervisor: Prof. David Donghoon Hyeon

Projects

AI ProducerJul 2022 – Apr 2024

Developing technological custom music composition

at MARG, in collaboration with Pozalabs and Supertone Inc.

- Developed a comprehensive music creation framework for musicians that generates new music based on reference audio.
- Served as project manager (Jul 2023 – Apr 2024)

Enhancing Cognitive Health through Optimal Hearing (ECHOH)Oct 2024 – Aug 2025

Global convergent research on healthcare solution to overcome hearing loss and dementia

at MARG, in collaboration with Ulsan Univ., KIST, Bell Therapeutics, Ewha Womans Univ., and Samsung Medical Center

- Conducted research on EEG-conditioned attended speech extraction.

Talks

7th SNU Brown Bag SeminarMar 2026

- Oral presentation on the paper *MGE-LDM: Joint Latent Diffusion for Simultaneous Music Generation and Source Extraction*

The First Korean Society for Music Informatics Conference (KSMI)Mar 2026

- Global Research Track
- Oral presentation on the paper *MGE-LDM: Joint Latent Diffusion for Simultaneous Music Generation and Source Extraction*

Others

Poster presentation at Music and Audio Workshop, Seoul (homepage) – Exploiting TF-Conformers for Music Audio Enhancement – Self-refining of Pseudo Labels for Music Source Separation with Noisy Labeled Data	Jul 2023
Oral Presentation at Sound Demixing Workshop, Milan (homepage), (abstract) – Self-refining of Pseudo Labels for Music Source Separation with Noisy Labeled Data	Nov 2023
Poster presentation at Korean Society for Music Informatics (homepage) – Lyrics Generation with Song form-aware Syllable Count Control	Apr 2025
Military Service <i>Korean Army, Honorable Discharge</i>	May 2016 – Feb 2018

Peer Reviewer

NeurIPS 2025 Workshop on Multimodal Algorithmic Reasoning
NeurIPS 2025 Workshop on AI for Music Committee
ICASSP 2026
INTERSPEECH 2026
KSMI 2026
IEEE SLT 2026
NeurIPS 2026
NeurIPS 2026 AI4Music Workshop Committee

Skills

Languages: Korean (Native), English (Professional Working), Japanese (Professional Working)
Programming: Python, PyTorch, PyTorch Lightning, NumPy, SciPy, Pandas, Scikit-learn
Frameworks/Tools: Jupyter, Git, Docker, Singularity